

3.

$$\vec{v} \in \mathbb{C} \Rightarrow \vec{v} = \begin{pmatrix} a+ib \\ x+iy \end{pmatrix} = \begin{pmatrix} a \\ x \end{pmatrix} + i \begin{pmatrix} b \\ y \end{pmatrix}$$

$$\operatorname{Re}\{\vec{v}\} = \begin{pmatrix} a \\ x \end{pmatrix} \quad \operatorname{Im}\{\vec{v}\} = \begin{pmatrix} b \\ y \end{pmatrix}$$

Beispiel

$$\vec{v}_1 = \begin{pmatrix} 1+i \\ 3-2i \end{pmatrix} \quad \vec{v}_2 = \begin{pmatrix} 10+8i \\ 5+i\sqrt{2} \end{pmatrix} \Rightarrow \mathbb{C}^2$$

$$\operatorname{Re}\{\vec{v}_1\} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} \quad \operatorname{Im}\{\vec{v}_1\} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$\operatorname{Re}\{\vec{v}_2\} = \begin{pmatrix} 10 \\ 5 \end{pmatrix} \quad \operatorname{Im}\{\vec{v}_2\} = \begin{pmatrix} 8 \\ \sqrt{2} \end{pmatrix}$$

$$\vec{v}_1 + \vec{v}_2 = \begin{pmatrix} 1+i+10+8i \\ 3-2i+5+i\sqrt{2} \end{pmatrix} = \begin{pmatrix} 11+9i \\ 8+i(\sqrt{2}-2) \end{pmatrix}$$

$$\alpha = i\sqrt{3}$$

$$\vec{v}_3 = \alpha \vec{v}_1 = i\sqrt{3} \begin{pmatrix} 1+i \\ 3-2i \end{pmatrix} = \begin{pmatrix} i\sqrt{3} - \sqrt{3} \\ 3\sqrt{3}i + 2\sqrt{3} \end{pmatrix}$$

$$5) \quad \vec{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad u_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad u_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad u_3 = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

$$\vec{v} = \alpha u_1 + \beta u_2 + \gamma u_3 \quad \text{mit } \alpha, \beta, \gamma \in \mathbb{R}$$

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \gamma \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} \alpha + \beta + \gamma \\ \alpha - \beta - 3\gamma \end{pmatrix} \Rightarrow \begin{cases} 1 = \alpha + \beta + \gamma \\ 2 = \alpha - \beta - 3\gamma \end{cases}$$

System der
Gleichungen

$$\overbrace{\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -3 \end{pmatrix}}^A \overbrace{\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}}^x = \overbrace{\begin{pmatrix} 1 \\ 2 \end{pmatrix}}^b \quad \text{sistema} \Rightarrow Ax = b$$

sistemas Incompatible (no existe solución)

compatible indeterminado (tiene infinitas soluciones)

compatible determinado (tiene una única solución)

$$\begin{cases} \alpha + \beta + \gamma = 1 & (1) \\ \alpha - \beta - 3\gamma = 2 & (2) \end{cases} \Rightarrow -\alpha = \beta + 3\gamma + 2$$

reemplazamos en (1) $\Rightarrow \beta + 3\gamma + 2 + \beta + \gamma = 1$

$$2\beta + 4\gamma = -1 \Rightarrow \beta + 2\gamma = -\frac{1}{2}$$

$$\alpha = \beta + 3\gamma + 2 = -\frac{1}{2} - 2\gamma + 3\gamma + 2 = \frac{3}{2} + \gamma$$

$$\beta = -\frac{1}{2} - 2\gamma$$

$$\beta = -\frac{1}{2} - 2\gamma$$

$$\text{sol: } \begin{pmatrix} \frac{3}{2} + \gamma \\ -\frac{1}{2} - 2\gamma \\ \gamma \end{pmatrix}$$

sol particular del sistema

$$\Rightarrow X = \underbrace{\begin{pmatrix} \frac{3}{2} \\ -\frac{1}{2} \\ 0 \end{pmatrix}}_{x_p} + \underbrace{\begin{pmatrix} \gamma \\ -2\gamma \\ \gamma \end{pmatrix}}_{x_h}$$

sol del homogéneo

obs sistema homogéneo

$$A\vec{x} = 0$$

$$\begin{cases} \alpha + \beta + \gamma = 0 & (1) \\ \alpha - \beta - 3\gamma = 0 & (2) \end{cases}$$

$$(1) + (2) \Rightarrow 2\alpha - 2\gamma = 0 \Rightarrow \alpha = \gamma$$

$$(1) - (2) \Rightarrow 2\beta + 4\gamma = 0 \Rightarrow \beta = -2\gamma$$

sol del homogéneo

$$\begin{pmatrix} \gamma \\ -2\gamma \\ \gamma \end{pmatrix}$$

verificar que x_p es sol. del sistema

$$A x_p = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -3 \end{pmatrix} \begin{pmatrix} 3/2 \\ -1/2 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{3}{2} - \frac{1}{2} + 0 \\ \frac{3}{2} + \frac{1}{2} + 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \checkmark$$

test

$$r = 1$$

$$p = -1/2 - 2 = -5/2$$

$$\alpha = 3/2 + 1 = 5/2$$

$$\frac{5}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \frac{5}{2} \begin{pmatrix} 0 \\ -1 \end{pmatrix} + \begin{pmatrix} 1 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

— 0 — 0 —

$$\begin{cases} \alpha + p + r = 1 \\ \alpha - p - 3r = 2 \end{cases} \Rightarrow A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -3 \end{pmatrix} \quad b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & -1 & -3 & 2 \end{array} \right) \xrightarrow{R_2 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -2 & -4 & 1 \end{array} \right) \xrightarrow{\frac{1}{2}R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -1 & -2 & \frac{1}{2} \end{array} \right)$$

$$\begin{cases} \alpha + p + r = 1 \\ -p - 2r = \frac{1}{2} \end{cases}$$

— 0 — 0 —

$$7) \quad v = \begin{pmatrix} 2 \\ -5 \\ 3 \end{pmatrix} \quad w_1 = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} \quad w_2 = \begin{pmatrix} 2 \\ -4 \\ -1 \end{pmatrix} \quad w_3 = \begin{pmatrix} 1 \\ -5 \\ 1 \end{pmatrix}$$

$$v = \alpha w_1 + p w_2 + r w_3$$

$$\begin{cases} \alpha + 2p + r = 2 \\ -3\alpha - 4p - 5r = -5 \\ 3\alpha + 2p - r = 1 \end{cases} \Rightarrow A = \begin{pmatrix} 1 & 2 & 1 \\ -3 & -4 & -5 \\ 3 & 2 & -1 \end{pmatrix} \quad b = \begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix}$$

$$x = \begin{pmatrix} \alpha \\ p \\ r \end{pmatrix}$$

$$A X = b$$

$$\text{si existe } A^{-1} \Rightarrow \underbrace{A^{-1}A}_I X = A^{-1}b \Rightarrow X = A^{-1}b$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ -3 & -4 & -5 & 0 & 1 & 0 \\ 2 & -1 & 1 & 0 & 0 & 1 \end{array} \right) \xrightarrow[\substack{F_2 + 3F_1 \\ F_3 - 2F_1}]{F_3 - 2F_1} \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 2 & -2 & 3 & 1 & 0 \\ 0 & -5 & -1 & -2 & 0 & 1 \end{array} \right)$$

$$\xrightarrow{F_3 + \frac{5}{2}F_2} \left(\begin{array}{ccc|ccc} & & & & & \end{array} \right)$$

triangulando el sistema

$$A X = b \quad \left(\begin{array}{ccc} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{array} \right) \left(\begin{array}{c} \alpha \\ \beta \\ \gamma \end{array} \right) = \left(\begin{array}{c} 2 \\ -5 \\ 1 \end{array} \right)$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 2 & & \\ -3 & -4 & -5 & -5 & & \\ 2 & -1 & 1 & 1 & & \end{array} \right) \xrightarrow[\substack{F_3 - 2F_1 \\ F_2 + 3F_1}]{F_2 + 3F_1} \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 2 & & \\ 0 & 2 & -2 & 1 & & \\ 0 & -5 & -1 & -3 & & \end{array} \right) \xrightarrow{F_3 + \frac{5}{2}F_2}$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 2 & & \\ 0 & 2 & -2 & 1 & & \\ 0 & 0 & -6 & -\frac{1}{2} & & \end{array} \right)$$

$$\left\{ \begin{array}{l} \alpha + 2\beta + \gamma = 2 \\ 2\beta - 2\gamma = 1 \\ -\gamma \cdot 6 = -\frac{1}{2} \Rightarrow \gamma = \frac{1}{12} \end{array} \right.$$

$$\left\{ \begin{array}{l} \alpha + 2\beta + \frac{1}{12} = 2 \\ 2\beta - \frac{1}{6} = 1 \Rightarrow \beta = \left(1 + \frac{1}{6}\right) \frac{1}{2} \end{array} \right.$$

3.d) $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbb{R}^{2 \times 2}$

$$A^2 = A \quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$A^2 = A \cdot A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^2 + bc & ab + bd \\ ca + dc & cb + d^2 \end{pmatrix}$$

$$\begin{cases} a^2 + bc = a \\ b(a+d) = b \\ c(a+d) = c \\ d^2 + bc = d \end{cases} \quad \leftarrow \quad A^2 = A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\begin{cases} a^2 + bc = a \\ b(a+d-1) = 0 \\ c(a+d-1) = 0 \\ d^2 + bc = d \end{cases}$$

$\rightarrow$ si $b=0 \wedge c=0$

$$\begin{cases} a^2 = a \\ d^2 = d \end{cases} \Rightarrow \begin{cases} a(a-1) = 0 \\ d(d-1) = 0 \end{cases}$$

sol: $a=0 \Rightarrow b=0 \quad c=0 \quad d=0$

$a=1 \quad b=0 \quad c=0 \quad d=1$

$a=0 \quad b=0 \quad c=0 \quad d=1$

$a=1 \quad b=0 \quad c=0 \quad d=0$

$$\Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}; \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}; \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

si $a+d-1=0 \Rightarrow d=1-a$

$$\begin{cases} a^2 + bc = a \end{cases} \Rightarrow a^2 - a + bc = 0 \quad \leftarrow$$

$$\begin{cases} (1-a)^2 + bc = 1-a \end{cases} \Rightarrow a^2 - 2a + 1 + bc - 1 + a = 0$$

$$a^2 - a + bc = 0 \quad \leftarrow$$

$$\boxed{a^2 - a + bc = 0}$$

$$a(a-1) + bc = 0$$

$$\begin{pmatrix} a & b \\ c & 1-a \end{pmatrix}$$